

Rosen — §2.3: Functions

Selected Exercises

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Exercise 27

- (a) Prove that a strictly decreasing function from $\mathbb{R}$ to itself is one-to-one.
- (b) Give an example of a decreasing function from $\mathbb{R}$ to itself that is not one-to-one.

Exercise 35

If f and $f \circ g$ are onto, does it follow that g is onto? Justify your answer.

Exercise 41

Recall (Exercise 40): if f is a function from A to B and S, T are subsets of A , then $f(S \cap T) \subseteq f(S) \cap f(T)$.

- (a) Give an example to show that the inclusion above may be proper.
- (b) Show that if f is one-to-one, the inclusion above is an equality.

Exercise 45

Let f be a function from A to B . Let S be a subset of B . Show that $f^{-1}(\overline{S}) = \overline{f^{-1}(S)}$.

Exercise 71

Let S be a subset of a universal set U . The characteristic function f_S of S is the function from U to the set $\{0, 1\}$ such that $f_S(x) = 1$ if x belongs to S and $f_S(x) = 0$ if x does not belong to S . Let A and B be sets. Show that for all $x \in U$,

- (a) $f_{A \cap B}(x) = f_A(x) \cdot f_B(x)$
- (b) $f_{A \cup B}(x) = f_A(x) + f_B(x) - f_A(x) \cdot f_B(x)$
- (c) $f_{\overline{A}}(x) = 1 - f_A(x)$
- (d) $f_{A \oplus B}(x) = f_A(x) + f_B(x) - 2f_A(x)f_B(x)$

Exercise 75

Prove that if x is a positive real number, then

- (a) $\lfloor \sqrt{\lfloor x \rfloor} \rfloor = \lfloor \sqrt{x} \rfloor$.

(b) $\lceil \sqrt{\lceil x \rceil} \rceil = \lceil \sqrt{x} \rceil$.

Exercise 77

For each of these partial functions, determine its domain, codomain, domain of definition, and the set of values for which it is undefined. Also, determine whether it is a total function.

(a) $f : \mathbb{Z} \rightarrow \mathbb{R}, f(n) = 1/n$

(b) $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(n) = \lceil n/2 \rceil$

(c) $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Q}, f(m, n) = m/n$

(d) $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}, f(m, n) = mn$

(e) $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}, f(m, n) = m - n$ if $m > n$

Exercise 78

- (a) Show that a partial function from A to B can be viewed as a function f^* from A to $B \cup \{u\}$, where u is not an element of B and

$$f^*(a) = \begin{cases} f(a) & \text{if } a \text{ belongs to the domain of definition of } f \\ u & \text{if } f \text{ is undefined at } a. \end{cases}$$

- (b) Using the construction in (a), find the function f^* corresponding to each partial function in Exercise 77.

Exercise 79

- (a) Show that if a set S has cardinality m , where m is a positive integer, then there is a one-to-one correspondence between S and the set $\{1, 2, \dots, m\}$.
- (b) Show that if S and T are two sets each with m elements, where m is a positive integer, then there is a one-to-one correspondence between S and T .

Exercise 80

Show that a set S is infinite if and only if there is a proper subset A of S such that there is a one-to-one correspondence between A and S .